

Reza Karimzadeh

PH.D., COMPUTER SCIENCE, COPENHAGEN UNIVERSITY

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SUMMARY

Innovative and results-driven Machine Learning Engineer and PhD candidate with experience building and deploying end-to-end deep learning systems across computer vision and medical imaging. Strong background in object detection, representation learning, 3D segmentation, and robust model generalization, with industry and multi-center collaboration experience. Comfortable owning the full ML lifecycle from data engineering to deployment. Currently reading evolutionary biology out of curiosity about complex adaptive systems beyond machine learning.

PROFESSIONAL EXPERIENCE

Machine Learning Intern | Research Intern

Amsterdam UMC | VU Amsterdam, Netherlands

January 2026 - Present

- Developed and validated a deep learning model (Siamese) to predict cancer treatment response using pre and post-treatment CT imaging data (n=502 patients).
- Built end-to-end pipeline (data preprocessing, training, cross-validation) using PyTorch and MONAI.
- Collaborated with oncologists and radiologists to ensure clinical relevance and interpretability.
- Led a cross-institutional project between University and Hospital.

Machine Learning Intern | Research Intern (Remote)

Geneva University Hospital, Geneva, Switzerland

September 2025 - Present

- Built a multi-center deep learning system to predict cancer subtype from PET scans by modeling each patient as a variable-length set of detected regions (151 training / 80 external cases).
- Implemented and benchmarked set-based architectures (Transformer, Deep Sets, MIL, GNN), achieved ROC-AUC up to 0.91 in cross-validation.
- Improved cross-site robustness using test time adaptation (entropy minimization), increasing external ROC-AUC from 0.81 to 0.87 without target labels.
- Led a fully remote university-hospital collaboration.

Deep Learning Engineer

PART Artificial Intelligence Startup

October 2022 - January 2023

- Developed and deployed YOLO-based object detection models for license plate detection and signature localization in official documents, optimizing performance for real-world production environments.
- Achieved mAP@0.5 of 0.97 for license plate detection and 0.94 for signature localization, improving detection accuracy and robustness under varying lighting, resolution, and quality conditions.
- Deployed with docker into a production pipeline with real-time inference and scalable API.

Technical Lead | Computer Vision & Signal Processing Engineer

Iran's National Elites Foundation

September 2020 - September 2021

- Served as Computer Vision Group Leader, leading development of a non-contact Heart Rate (HR) estimation system from RGB video (3.02 MAE for HR).
- Designed signal processing and feature extraction pipelines (ROI tracking, color analysis, temporal filtering, amplitude normalization) to improve signal robustness under motion and lighting variation.
- Coordinated deployment on mobile/embedded platforms for real-time inference.

Engineering Internship

Electro-Xray Company

July 2017 - September 2017

- Repair and maintenance of imaging systems (CT, Portable Radiology device, C-Arm, Mammography and OPG)

Teaching Assistant

Copenhagen University | Sharif University of Technology | Amirkabir University of Technology

- Deep Learning Course | Numerical Methods
- Medical Images Analysis and Processing | Image Processing | Medical Imaging Systems

EDUCATION & PAPERS	Ph.D. in Computer Science, Copenhagen, Denmark <i>Image Section, DIKU, Copenhagen University</i> <i>May 2023 - May 2026 (Expected)</i> Thesis: Applications of Deep Learning Algorithms for Medical Data Analysis - A novel Explainable AI method based on Gradient-driven contours (Link) - Contrastive Learning Framework for 3D Patch-Based Medical Image Segmentation (Link) - Generating synthetic CT scans from MRI images for radiotherapy (Link)	
	M.Sc. Degree in Biomedical Engineering, Tehran, Iran <i>Sharif University of Technology</i> <i>Sep. 2019 - May 2022</i> Thesis: Medical Image Segmentation Using Deep Learning Methods - Attention-based deep learning segmentation: Application to brain tumor delineation (Link) - Proposing a shape-based loss function for organ segmentation in 3D medical images (Link) - COVID-19 detection from the chest CT scans (Link)	
	B.Sc. Degree in Biomedical Engineering, Tehran, Iran <i>Amirkabir University of Technology</i> <i>Sep. 2014 - Sep. 2019</i> Thesis: Design and Implementation of Brain Surgery Bipolar Electrocautery Simulator Using Haptic Technology (Link)	
TECHNICAL SKILLS	Programming Frameworks DL & ML Methods MLOps CI/CD Systems Documentation Soft Skills Languages	Python, MATLAB PyTorch, MONAI, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib CNNs, Transformers, SSL, XAI, Object Detection, Domain Adaptation Git, GitHub, ONNX, Docker, Linux L ^A T _E X, Markdown, Microsoft Office Communication, Leadership, Problem-Solving, Adaptability, Teamwork English, Persian, Danish (basic)
OPEN SOURCE PROJECTS	Extremal Contours <i>Copenhagen University</i> GitHub - Developed an XAI method for extracting the most informative part of image using Fourier Transform	
	Self-supervised Learning Framework for 3D Patch-based Image Segmentation <i>Copenhagen University</i> GitHub - Implemented a self-supervised learning framework for 3D patch-based medical image segmentation	
	A Non-contact heart rate estimation framework <i>Iran's National Elites Foundation</i> GitHub - Developed a non-contact heart rate estimation system from facial video sequences	
LEADERSHIP & PROFESSIONAL SERVICE	Infrastructure Coordinator (GitHub, Slack, Trello) & Symposium Organization <i>Image Section, DIKU, University of Copenhagen</i> <i>Sep. 2023 - Present</i> - Managed shared research infrastructure and collaboration workflows - Co-organized quarterly scientific symposia, coordinating speakers, scheduling, and session structure	
	Workshop Tutor/Organizer <i>Advanced Health Technologies-Artificial Intelligence in Medicine</i> <i>December 2021</i> - Deep Learning workshop tutor and organizer (Link)	
	Peer Review Service <i>Reviewer for international journals and conferences</i> <i>2023 - Present</i> - Journal of Biomedical and Health Informatics Journal of Medical Imaging Scandinavian Conference on Image Analysis NeurIPS'24 AIM-FM Workshop	